



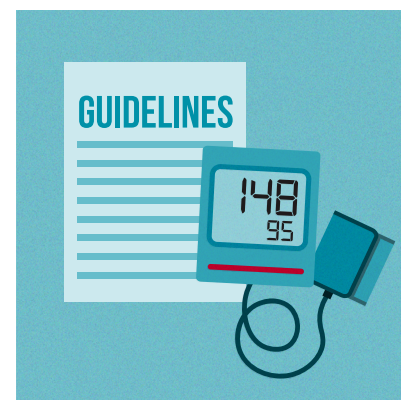
2017 ACC/AHA Hypertension Guideline: Highlights & Common Questions

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The 2017 American College of Cardiology (ACC)/American Heart Association (AHA) guidelines established lower blood pressure (BP) targets than previously recommended, and these targets have remained in effect since then. Below, we provide a summary of the recommendations, followed by common questions and answers related to this guideline.



Basic Changes to Hypertension Management Recommended by the 2017 ACC/AHA Hypertension Guideline¹

- 1. The most prominent update was the reduction in recommended BP levels that prompt the initiation of drug treatment for elevated BP and the BP goal in those requiring treatment.**
 - From 140/90 mmHg to 130/80 mmHg in those <60 years old and
 - From 150/90 mmHg to 130/80 mmHg in those ≥60.
- 2. Lower than 130/80 mmHg is the BP level used to define the level in nearly all clinical settings for initiating drug therapy* and defines the recommended BP target.**
- 3. A 10-year risk of cardiovascular disease threshold of 10%, as well as BP level, is used to determine patients who need to be treated with BP medications (in addition to lifestyle management).**
 - If atherosclerotic cardiovascular disease (ASCVD) risk > 10%, initiate drug treatment when systolic blood pressure (SBP) ≥ 130 mmHg or diastolic blood pressure (DBP) ≥ 80 mmHg.
 - If ASCVD risk < 10%, initiate drug treatment at SBP ≥ 140 mmHg or DBP ≥ 90 mmHg.
- 4. There is greater reliance on out-of-office BP levels for both the diagnosis and management of hypertension. (It has become increasingly recognized that we should not depend only on the measurement of BP in the office to manage hypertension.)**
 - Either ambulatory blood pressure monitoring (ABPM) or home blood pressure monitoring (HBPM) is needed to confirm diagnosis of hypertension requiring antihypertensive drug treatment and to confirm control in those on treatment, to detect white coat and masked hypertension.

5. The implications of the 2017 guideline on hypertension treatment are as follows.

- Based on the new threshold of 130/80 mmHg or greater, compared with the 140/90 mmHg or above threshold, the number of patients with hypertension diagnosis was expected to increase by 13.7%, to 45.6%, or an additional 31.1 million people nationally.¹⁻²
- Providers were advised to arrange for patient instruction on the use of home BP monitoring.
- Because of the requirement to risk stratify:
 - The number of newly identified hypertensive patients requiring treatment would only increase by 1.9% or 4.2 million.
 - However, medications would need to be intensified in an additional 14.4% of patients, or 7.9 million people, in those currently on antihypertensive medications.
 - Unlike the 2014 hypertension guideline published in JAMA (JNC-8) and the American Academy of Family Physicians (AAFP)/American College of Physicians (ACP) guideline, patients recommended for drug treatment are more likely be those at greatest risk and, therefore, most likely to benefit from treatment.
- Greater use of additional and more potent antihypertensive medications (e.g., chlorthalidone to replace hydrochlorothiazide and spironolactone) are needed to achieve the lower BP target (< 130/80 mmHg), as more patients were classified as resistant.
- There is a need for greater emphasis on lifestyle modification.
- Compared to achieving the JNC-8 treatment goals, decrease estimates for achieving the 2017 ACC/AHA guideline treatment goals are a decrease of 340,000 in cardiovascular disease (CVD) events and 157,000 fewer deaths in the United States, annually.³

* In patients with 10-year ASCVD risk < 10% or to prevent recurrent strokes in patients after an ischemic stroke, drug treatment should be initiated at $\geq$ 140/90 mmHg.

References

1. Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Hypertension*. 2018;71(6):1269-1324. doi:10.1161/HYP.0000000000000066.
2. Muntner P, Carey RM, Gidding S, et al. Potential US population impact of the 2017 ACC/AHA high blood pressure guideline. *Circulation*. 2018;137(2):109-118. doi:10.1161/CIRCULATIONAHA.117.032582.
3. Bundy JD, Mills KT, Chen J, et al. Estimating the association of the 2017 and 2014 hypertension guidelines with cardiovascular events and deaths in US adults: an analysis of national data. *JAMA Cardiol*. 2018;3(7):572-581. doi:10.1001/jamacardio.2018.1240.

Response to Questions and Common Criticisms

1. Does labeling patients as hypertensive have adverse health effects and increase drug use?

- This concern has been around for some time. It was discussed in 2003 when missed days at work as a result of hypertension labeling was raised by Haynes et al¹ during the development of JNC-7.
- This created significant initial concern about the term “prehypertension” introduced by that guideline to identify patients with BP 120-139 mmHg/80-89 mmHg and was intended to be used as an incentive to increase the use of lifestyle change to reduce the estimated 90% incidence of patients in that range who convert from prehypertension to sustained hypertension over 10-20 years.^{2,3}
- The success of the prehypertension label in stimulating lifestyle change has been disappointing.
- One hope for the stronger labeling (stage 1 hypertension) in low risk patients (ASCVD 10-year risk < 10% with SBP 130-140 mmHg) is to encourage greater attention to the need for lifestyle changes to reduce the need for antihypertensive drug treatment.

2. The emphasis of the guideline on a risk-based treatment strategy increases complexity and is without an evidence base.

- Patients with SBP 130-139 mmHg but CVD risk < 10% are recommended for lifestyle changes (as in those with previous prehypertensive designation but with a stronger label).
- Those at higher risk (with BP level only one factor) would be recommended for drug treatment (this is the population that, in addition to elevated CVD risk, has been the focus of recruitment into clinical outcome trials and demonstrated the most consistent benefit from drug treatment).
- Of note, in contrast to the 2014 guideline, younger patients are more likely to be recommended for lifestyle change, whereas, older patients (who are much more likely to be at risk of hypertension complications) are more likely to be recommended for drug treatment.

3. Blood pressure in the Systolic Blood Pressure Intervention Trial (SPRINT) was measured under idealized research conditions in the participating clinical sites, with the patient resting quietly and not doing anything for five minutes.

- The measurement procedure in SPRINT was the same as recommended by guidelines going back decades.
- Use of oscillometric monitors to measure BP has been the standard practice in hypertension drug treatment trials for more than 20 years.⁴
- Recent data on BP measurement from SPRINT and confirmed by a study from German investigators demonstrate little difference as to whether staff are present in the room or not as long as other recommended procedures for BP measurement in place.^{5,6}
- Thus, inadequate rest period, improper cuff size, use of a non-validated BP monitor, and poor patient and arm positioning (seated in a chair versus on an exam table) introduce more error than staff presence or absence.
- It is distressing to admit that the goal of < 130 mmHg rather than < 120 mmHg was recommended in part to compensate for the poor quality of BP measurements currently in practice in clinical settings.

4. There is a lack of evidence as to the benefit of the < 130/80 mmHg target.

- One of four pre-specified questions posed for systematic review prior to drafting the guideline was “What is the optimal target for BP lowering during antihypertensive therapy in adults?” The evidence for this recommendation was based on this systematic review, which was published separately.⁷
- It is supported by multiple meta-analyses published before and after inclusion of results from SPRINT.⁸⁻¹¹
- While no level of evidence is immune from criticism, the evidence supporting 130/80 mmHg was unanimously agreed upon by the multidisciplinary panel of authors and the organizations represented. (Note: ACP and AAFP were invited but refused participation in drafting the 2017 ACC/AHA guideline.)
- It is worth noting that the < 150/90 mmHg treatment recommendation for patients over age 60 in the JAMA 2014 guideline was decided by only a single vote margin.¹²

5. Harms: The risk associated with the lower BP target is excessive compared to the benefit, especially in older patients (age 60 and over).

- Critics like to suggest that the modest risk (< 1-2%) of hypotension, syncope, even acute kidney injury (AKI) (95% of which either totally or partially resolves [90% show complete resolution])¹³ should be considered equivalent to myocardial infarction (MI), stroke, acute decompensated heart failure, and cardiovascular death.^{14,15}
- Importantly, no difference in rates of serious adverse events overall or in the above selected adverse events (except for AKI) were seen in older patients (even in frail though ambulatory patients over age 75) compared to overall population.¹⁶⁻¹⁹
- Patients older than 75 saw the greatest overall benefit in terms of CVD, cardiovascular mortality and all-cause mortality reduction compared to younger patients (though age X treatment interaction was not significant). Of note, the number of patients needed to treat to prevent a CVD event was 61 overall in SPRINT but only 28 in patients age ≥ 75 needed. For all-cause mortality, the numbers were 90 and 41, respectively.^{16, 17}

6. Evidence for the < 130/80 mmHg target in a person with diabetes based on the results in the ACCORD trial is unconvincing.

- ACCORD was half the size of SPRINT, with a 3-way factorial design, thus markedly underpowered.
- Recommendation for the < 130/80 target in a person with diabetes is based on a pre-specified, systematic review of trials, that included ACCORD, prior to drafting the guideline.⁷
- Meta-analysis of SPRINT and ACCORD show overlap in findings,²⁰ and SPRINT results in participants with prediabetes and metabolic syndrome show reduction in CVD and mortality similar to the overall trial results.^{21,22}
- Since publication of the guidelines, additional evidence supporting tighter control in people with diabetes has been published.²³

7. It is unclear how relevant these results are to the millions of younger adults who have been newly labeled with hypertension based on the new guidelines.

- No clinical outcome trial data has ever been available to select the optimal BP threshold or treatment target in younger patients with hypertension.
- Clinical event rate is too low in this population to assess effect of treatment in a clinical outcome trial.

8. Feasibility of implementing the BP recommendation is a challenge.

- While feasibility was considered, the major focus of the panel was to base recommendations on the best evidence for benefit of the patient (not the provider).
- While the challenges were recognized (and that they would require overdue changes to the way this deadly, disabling, costly disease is managed), none were felt to be unsurmountable.

9. Evidence for DBP targets is less convincing.

- This evidence gap is acknowledged by the lower level of evidence designated for the DBP recommendations.
- However, poor control of DBP is rarely the cause for classifying patients at the highest risk (e.g., > age 55, other cardiovascular risk factors, and clinical/subclinical CVD) as out of control.

References

1. Haynes RB, Sackett DL, Taylor DW, et al. Increased absenteeism from work after detection and labeling of hypertensive patients. *N Engl J Med*. 1978;299(14):741-744. doi:10.1056/NEJM197810052991403.
2. Vasan RS, Larson MG, Leip EP, et al. Impact of high-normal blood pressure on the risk of cardiovascular disease. *N Engl J Med*. 2001;345(18):1291-1297. doi:10.1056/NEJMoa003417.
3. Chobanian AV, Bakris GL, Black HR, et al. The seventh report of the Joint National Committee on prevention, detection, evaluation, and treatment of high blood pressure: the JNC 7 report. *JAMA*. 2003;289(19):2560-2572. doi:10.1001/jama.289.19.2560.
4. Giorgini P, Weder AB, Jackson EA, Brook RD. A review of blood pressure measurement protocols among hypertension trials: implications for "evidence-based" clinical practice. *J Am Soc Hypertens*. 2014;8(9):670-676. doi:10.1016/j.jash.2014.07.024.
5. Johnson KC, Whelton PK, Cushman WC, et al. Blood pressure measurement in SPRINT (Systolic Blood Pressure Intervention Trial). *Hypertension*. 2018;71(5):848. doi:10.1161/HYPERTENSIONAHA.117.10479.
6. Bauer F, Seibert FS, Rohn B, et al. Attended versus unattended blood pressure measurement in a real life setting. *Hypertension*. 2018;71(2):243. doi:10.1161/HYPERTENSIONAHA.117.10026.
7. Reboussin DM, Allen NB, Griswold ME, et al. Systematic review for the 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2018;71(19):2176-2198. doi:10.1016/j.jacc.2017.11.004.
8. Bundy JD, Li C, Stuchlik P, et al. Systolic blood pressure reduction and risk of cardiovascular disease and mortality: a systematic review and network meta-analysis. *JAMA Cardiol*. 2017;2(7):775-781. doi:10.1001/jamacardio.2017.1421.
9. Xie X, Atkins E, Lv J, et al. Effects of intensive blood pressure lowering on cardiovascular and renal outcomes: updated systematic review and meta-analysis. *Lancet*. 2016;387(10017):435-443. doi:10.1016/S0140-6736(15)00805-3.
10. Bangalore S, Toklu B, Gianos E, et al. Optimal systolic blood pressure target after SPRINT: insights from a network meta-analysis of randomized trials. *Am J Med*. 2017;130(6):707-719.e8. doi:10.1016/j.amjmed.2017.01.004.
11. Verdecchia P, Angeli F, Gentile G, Reboldi G. More versus less intensive blood pressure-lowering strategy: cumulative evidence and trial sequential analysis. *Hypertension*. 2016;68(3):642-653. doi:10.1161/HYPERTENSIONAHA.116.07608.
12. Wright JT Jr, Fine LJ, Lackland DT, et al. Evidence supporting a systolic blood pressure goal of less than 150 mm Hg in patients aged 60 years or older: the minority view. *Ann Intern Med*. 2014;160(7):499-503. doi:10.7326/M13-2981.
13. Rocco MV, Sink KM, Lovato LC, et al. Effects of intensive blood pressure treatment on acute kidney injury events in the Systolic Blood Pressure Intervention Trial (SPRINT). *Am J Kidney Dis*. 2018;71(3):352-361. doi:10.1053/j.ajkd.2017.08.021.
14. Clinical Guidelines Committee of the American College of Physicians, Wilt TJ, Kansagara D, Qaseem A. Hypertension limbo: balancing benefits, harms, and patient preferences before we lower the bar on blood pressure. *Ann Intern Med*. 2018;168(5): 369-370. doi:10.7326/M17-3293.
15. Qaseem A, Wilt TJ, Rich R, et al. Pharmacologic treatment of hypertension in adults aged 60 years or older to higher versus lower blood pressure targets: a clinical practice guideline from the American College of Physicians and the American Academy of Family Physicians. *Ann Intern Med*. 2017;166(6):430-437. doi:10.7326/M16-1785.
16. SPRINT Research Group, Wright JT Jr, Williamson JD, et al. A randomized trial of intensive versus standard blood-pressure control. *N Engl J Med*. 2015;373(22):2103-2116. doi:10.1056/NEJMoa1511939.
17. Williamson JD, Supiano MA, Applegate WB, et al. Intensive vs standard blood pressure control and cardiovascular disease outcomes in adults aged ≥75 years: a randomized clinical trial. *JAMA*. 2016;315(24):2673-2682. doi:10.1001/jama.2016.7050.
18. Sink KM, Evans GW, Shorr RI, et al. Syncope, hypotension, and falls in the treatment of hypertension: results from the randomized clinical Systolic Blood Pressure Intervention Trial. *J Am Geriatr Soc*. 2018;66(4):679-686. doi:10.1111/jgs.15236.
19. Rich MW, Ouslander JG. Hypertension in older adults in the wake of the Systolic Blood Pressure Intervention Trial. *J Am Geriatr Soc*. 2018;66(4):652-654. doi.org/10.1111/jgs.15326.
20. Perkovic V, Rodgers A. Redefining blood-pressure targets--SPRINT starts the marathon. *N Engl J Med*. 2015;373(22):2175-2178. doi:10.1056/NEJMe1513301.
21. Bress AP, King JB, Kreider KE, et al. Effect of intensive versus standard blood pressure treatment according to baseline prediabetes status: a post hoc analysis of a randomized trial. *Diabetes Care*. 2017;40(10):1401-1408. doi:10.2337/dc17-0885.
22. Dungan K, Craven TE, Soe K, et al. Influence of metabolic syndrome and race on the relationship between intensive blood pressure control and cardiovascular outcomes in the SPRINT cohort. *Diabetes Obes Metab*. 2018;20(3):629-637. doi:10.1111/dom.13127.
23. Bi Y, Li M, Liu Y, et al. Intensive blood-pressure control in patients with type 2 diabetes. *N Engl J Med*. Published online November 16, 2024. doi:10.1056/NEJMoa2412006.

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